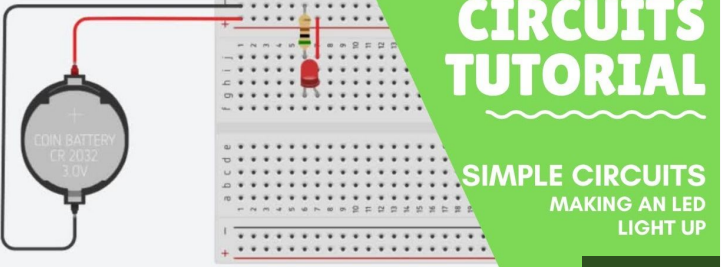


Lesson 07 — Tinkercad Circuits

Before we touch a single wire in the real world, let's play around in a **virtual sandbox**. Tinkercad Circuits lets us build and test electronics online — no burnt components, no wasted money. Today we'll explore the platform, wire up a simple LED circuit, and even make it fade using PWM. Let's get into it!

What is Tinkercad Circuits?



**TINKERCAD
CIRCUITS
TUTORIAL**

**SIMPLE CIRCUITS
MAKING AN LED
LIGHT UP**

▶ 09:57

 YouTube

Tinkercad Circuits Tutorial – Making a Simple LED

Learn how to use a breadboard in this beginner circuit class. Tinkercad Circuits is a great way to build and test circuits online before building them in real life. Take...



★ The Big Idea

Tinkercad Circuits is a **free, browser-based electronics simulator** by Autodesk. You can drag-and-drop components (Arduino, LEDs, resistors, motors, sensors), wire them up, write code, and hit **Simulate** to watch it run — all without buying a single physical part.

Think of it as a **flight simulator for electronics** 🛩️

Why Use It?

-  **Zero cost** — just a browser and internet
-  **Safe** — can't short-circuit or fry anything
-  **Beginner-friendly** — drag-and-drop interface
-  **Instant feedback** — see results immediately
-  **Shareable** — send your circuit link to friends
-  **Arduino built-in** — code editor included



Pro tip: Use Tinkercad to test your circuit *before* you build it physically. Saves time, money, and frustration!

The Basic Workflow

Here's the step-by-step process you'll follow every time you start a new circuit in Tinkercad. It's simple — promise!



Create Circuit

Log in → Circuits → "Create new Circuit"



Add Components

Drag Arduino, LED, resistor from the panel



Connect Wires

Click and drag wires between pins



Write Code

Open Code Editor, type Arduino sketch



Simulate & Test

Click Start Simulation, observe output

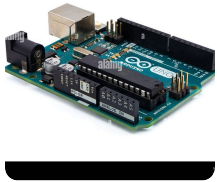


Remember: Always double-check your wiring *before* hitting Simulate. A wrong connection can "damage" virtual components too!

Components You'll Need



For our first circuit, we need just **4 components**. Here's a quick rundown of each one and what it does:



Arduino Uno

The **brain** of the circuit.
Runs your code and controls output pins. We use digital pin 9 for PWM output.



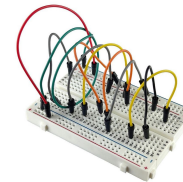
LED

Light Emitting Diode. Has a longer leg (anode +) and shorter leg (cathode -). Only lets current flow one way!



Resistor (220Ω)

Limits current flowing through the LED. Without it, the LED burns out! 220Ω is a safe value for 5V circuits.

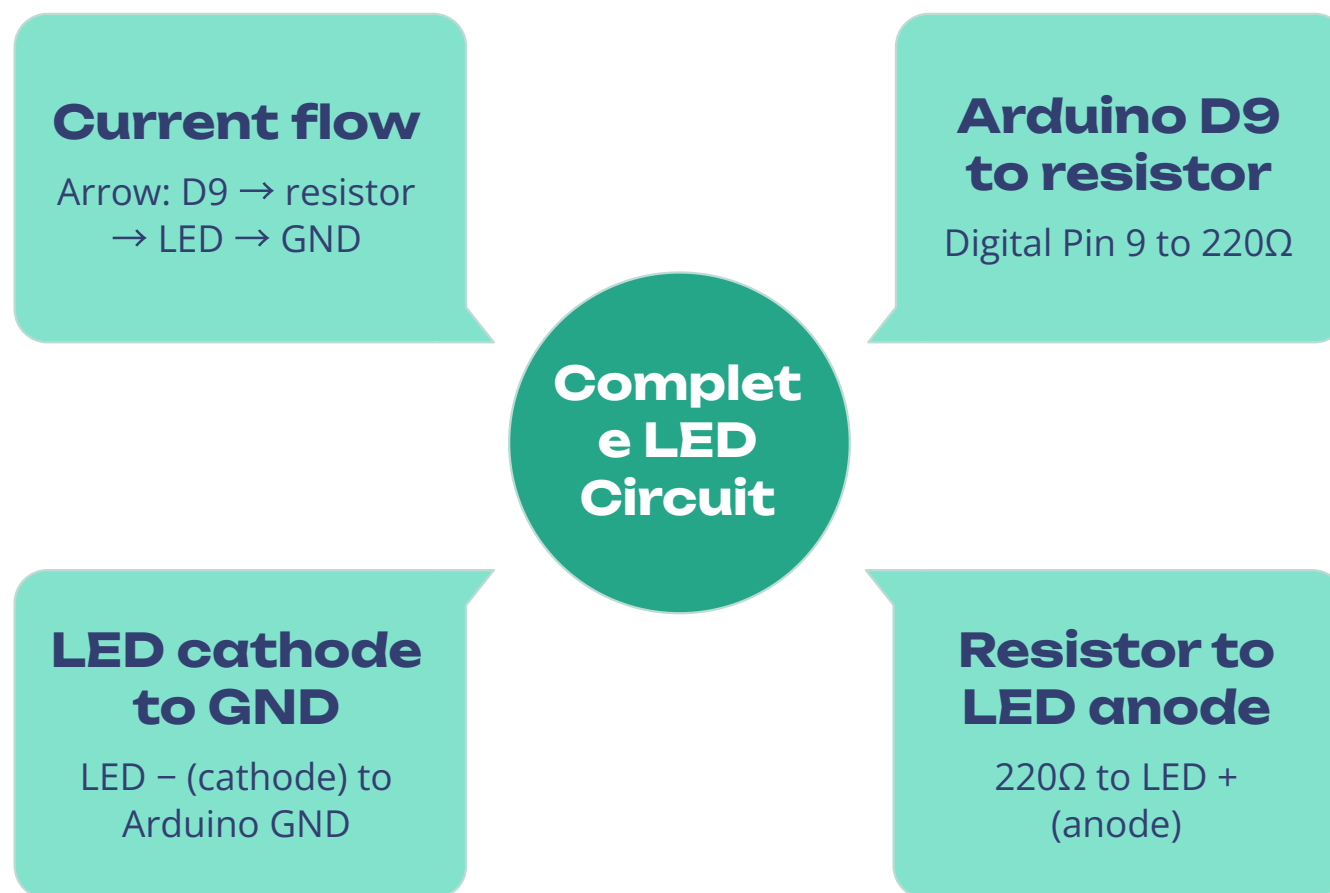


Breadboard & Wires

The **prototyping board** holds components without soldering. Red rail = +, Blue rail = GND (-).

Simple LED Circuit — Wiring Diagram

Here's how everything connects. Follow this exact wiring in Tinkercad:



The diagram above shows the complete signal path — from the Arduino's PWM output pin, through the current-limiting resistor, into the LED, and back to ground.

● Positive Side

- Arduino **Pin 9** → Resistor (220Ω)
- Resistor → LED **Anode (+)** (longer leg)

● Negative Side

- LED **Cathode (-)** (shorter leg) → GND
- GND → Arduino **GND pin**

 ⚡ **Key rule:** Always use a resistor with an LED! Direct connection to 5V will destroy it. The resistor **limits current** to a safe level (~20mA).

Arduino Code — Blink the LED

Once your circuit is wired, click "**Code**" in the top-right of Tinkercad. Switch to **Text** mode and type this:

```
// Lesson 07 — Simple LED Blink
// Pin 9 controls the LED

int ledPin = 9; // LED connected to digital pin 9

void setup() {
  pinMode(ledPin, OUTPUT); // Set pin as output
}

void loop() {
  digitalWrite(ledPin, HIGH); // LED ON
  delay(1000);                // Wait 1 second
  digitalWrite(ledPin, LOW);  // LED OFF
  delay(1000);                // Wait 1 second
}
```

Code Breakdown

- **int ledPin = 9** — names our output pin
- **setup()** — runs once, sets pin mode
- **loop()** — runs repeatedly
- **HIGH** = 5V (ON), **LOW** = 0V (OFF)
- **delay(1000)** = 1000 milliseconds = 1 second

How to Run

1. Click "**Start Simulation**" button
2. Watch the LED blink ON/OFF every second
3. Change `delay()` values to adjust speed
4. Try `delay(500)` — faster blink!

What is PWM?

Pulse Width Modulation

PWM is a clever trick to **simulate analog output** using only digital pins. The Arduino rapidly switches the pin ON and OFF. By changing *how long* it stays ON vs OFF (the **duty cycle**), we control average power — and thus LED brightness!

- **0% duty cycle** = always OFF = LED dark
- **50% duty cycle** = half ON = medium brightness
- **100% duty cycle** = always ON = full brightness



On Arduino, PWM values range from **0 to 255**. Only certain pins support PWM — marked with ~ (tilde), like pins 3, 5, 6, 9, 10, 11.

LED Fading with PWM

Now let's make the LED **smoothly fade in and out** instead of just blinking. We use `analogWrite()` with a `for` loop:

```
// Lesson 07 — LED Fade (PWM)
int ledPin = 9; // PWM pin (has ~ symbol)

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  // Fade IN — brightness 0 to 255
  for (int brightness = 0; brightness <= 255; brightness++) {
    analogWrite(ledPin, brightness);
    delay(10); // small delay for smooth fade
  }

  // Fade OUT — brightness 255 to 0
  for (int brightness = 255; brightness >= 0; brightness--) {
    analogWrite(ledPin, brightness);
    delay(10);
  }
}
```

`analogWrite(pin, value)`

Sends PWM signal. Value = **0–255**.
Pin must be a PWM pin (~)

`for` loop

Counts up (0→255) then down (255→0), changing brightness step by step

`delay(10)`

Small pause between steps.
Smaller = smoother but faster. Try 5 or 20!

Running the Simulation in Tinkercad



Here's a walkthrough of the Tinkercad interface when you're ready to test your circuit:

1

Build Your Circuit

Drag components from the left panel onto the workspace. Connect wires by clicking a pin and dragging to another pin. Red = positive, Black = GND.

2

Open Code Editor

Click the **"Code"** button (top-right). Choose **Text** mode (not Blocks). Paste or type your Arduino sketch.

3

Start Simulation

Click the big **"Start Simulation"** button. The LED should light up or fade based on your code. Watch the Serial Monitor if needed.

4

Debug & Iterate

If something's wrong — check wire colors, pin numbers, and code syntax. Stop simulation, fix, and re-run. This is the **engineering process!**

Practice Task — Your Turn!

Build an LED circuit in Tinkercad and make the LED fade gradually from dark to bright and back. Experiment with different delay values and observe the effect!

Checklist

- Create new Circuit in Tinkercad
- Add Arduino Uno + LED + 220Ω resistor
- Wire Pin 9 Resistor LED GND
- Write the fade code with `analogWrite()`
- Run simulation — LED should fade smoothly
- Try changing `delay()` to 5, 20, 50
- Bonus: Add a second LED on Pin 10!

Experiment Ideas

- What happens if you remove the resistor?
- Try two LEDs fading in opposite directions
- Change the for loop step: `brightness += 5`
- Use `millis()` instead of `delay()`
- Share your circuit link with a classmate!

 ★ **Next Lesson:** We'll add a potentiometer (knob) to control LED brightness in real-time using `analogRead()`!

5

Components Used

Arduino, LED, Resistor, Breadboard,
Wires

255

PWM Levels

Brightness values from 0 (off) to 255
(full)

6

PWM Pins

Pins 3, 5, 6, 9, 10, 11 support
`analogWrite()`